JEFFRY IRWAN GULTOM

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PROFESSIONAL SUMMARY

Computer Engineering graduate with a multidisciplinary technical foundation bridging embedded systems, IoT development, and software solutions, reinforced by extensive hands-on experience in industrial manufacturing environments. Demonstrated expertise in programming microcontrollers (ESP32/C++), developing responsive web interfaces, and maintaining electromechanical production machinery. Proven ability to translate complex technical requirements into functional prototypes, conduct rigorous system troubleshooting, and ensure operational precision. Adaptable and ready to contribute across embedded hardware engineering, industrial automation, or software-driven technical roles.

TECHNICAL SKILLS

Hardware & Embedded Systems: ESP32, Arduino, Microcontroller Architecture, Sensor & Actuator Interfacing, Serial Protocols (I2C, SPI, UART), Circuit Prototyping, Preventive Maintenance, Machine Tooling Setup

Programming & Software: C, C++, JavaScript, Python, HTML5, CSS3

Tools & Platforms: Git, GitHub, VS Code, Arduino IDE, IoT Cloud Platforms (Blynk, MQTT Broker), Microsoft Office Suite

Core Competencies: Hardware-Software Integration, Root Cause Analysis, Technical Documentation, Quality & Workplace Safety Compliance

TECHNICAL PROJECTS

IoT-Based Smart Port Berth & Environmental Monitoring System

Lead Developer

- Designed and built an end-to-end IoT telemetry prototype using an ESP32 microcontroller, integrating ultrasonic and liquid level sensors to track operational parameters in real time.
- Authored clean, modular firmware in C/C++ implementing non-blocking sensor acquisition algorithms and fault-handling routines.
- Configured wireless communication protocols to stream real-time sensor metrics to cloud IoT dashboards and local OLED displays, achieving sub-second data refresh intervals.
- Conducted hardware calibration, voltage stabilization, and rigorous system field testing to guarantee measurement accuracy.

Responsive Web-Based Data Management & Dashboard

Frontend Developer

- Engineered responsive, cross-platform user interfaces using HTML5, CSS3, and modern JavaScript to present dynamic system data cleanly.
- Optimized user interaction flows, structured modular CSS components, and debugged cross-browser rendering inconsistencies.
- Documented front-end architectural workflows and standardized UI component reusability for seamless deployment.

WORK & INTERNSHIP EXPERIENCE

Gamelab Indonesia — Frontend Developer Intern Sep 2025 – Jan 2026

- Built and optimized responsive web interfaces utilizing HTML5, CSS3, and JavaScript based on structured UI/UX specifications.
- Configured local development environments, resolved dependency version conflicts, and conducted functional cross-browser testing.
- Collaborated with team members across sprint cycles, delivering clean frontend code while meeting strict release deadlines.

Infinite Learning — Advanced AI Participant Aug 2024 – Jan 2025

- Studied core artificial intelligence architectures, machine learning workflows, and data preprocessing methodologies using Python.
- Implemented foundational classification and regression algorithms, analyzing data feature extraction techniques to support intelligent automation systems.

PT. Tai Cheng Development — Junior Technician Aug 2019 – Aug 2024

- Operated and maintained automated coil spring machinery within a high-volume manufacturing assembly line.
- Performed frontline troubleshooting, scheduled lubrication, and basic equipment repair to prevent recurring machine stoppages.
- Consistently achieved daily output quotas while adhering to rigorous corporate quality control protocols

PT. Satnusa Persada Tbk — Production Operator Aug 2017 – Feb 2019

- Executed precision tooling setups, changeovers, and calibration on heavy plastic molding machinery, strictly minimizing production downtime.
- Diagnosed electromechanical faults and performed preventive maintenance routines to maintain high-yield operational efficiency.
- Collaborated closely with production operators and engineering supervisors to enforce industrial safety (5S) and production quality standards.

EDUCATION

Institut Teknologi Batam (ITEBA) — Bachelor of Computer Engineering 2022 – 2026

SMK Widya Paramitra Batam — Technician Engineering 2014 – 2017